

trust the data provider to respond accurately to all queries. The third type of oracle is a custom, application-specific oracle service used by Maker and Compound. Its design differs based on the requirements of the protocol for which it was developed. For example, Compound relies on a single data provider that the Compound team controls to provide all on-chain price data to the Compound oracle.

Oracles, as they exist today, represent the highest risk to DeFi protocols that rely on them. All on-chain oracles are vulnerable to front-running, and millions of dollars have been lost due to arbitrageurs.¹⁴ Additionally, oracle services like Chainlink and Maker have suffered crippling outages with catastrophic downstream effects.¹⁵ Until oracles are blockchain native, hardened, and proven resilient, they represent the largest systemic threat to DeFi today.

SCALING RISK

As we have discussed, Ethereum and other proof-of-work (the consensus mechanism) blockchains have a fixed block size. For a block to become part of the chain, every Ethereum miner must execute all the included transactions on their machine. To expect each miner to process all the financial transactions for a global financial market is unrealistic. The current version of Ethereum is currently limited to a maximum of 30 transactions per second (TPS), yet almost all of DeFi today resides on this blockchain. Compared with Visa, which can handle upward of 65,000 TPS, Ethereum is